HOMEWORK 1 :

1.SUM OF TWO MATRICES:

```
#include <stdio.h>

int main()
{
    int rows, cols;

    printf("Enter number of rows: ");
    scanf("%d", &rows);

    printf("Enter number of columns: ");
    scanf("%d", &cols);

    int a[rows][cols];
    int b[rows][cols];
    int sum[rows][cols];

    printf("Enter first matrix:\n");
    for(int i = 0; i < rows; i++)
    {
        for(int j = 0; j < cols; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter second matrix:\n");
    for(int i = 0; i < rows; i++)
```

```
    {
        for(int j = 0; j < cols; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(int i = 0; i < rows; i++)
    {
        for(int j = 0; j < cols; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("Sum of matrices:\n");
    for(int i = 0; i < rows; i++)
    {
        for(int j = 0; j < cols; j++)
        {
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT :

Enter number of rows: 2

Enter number of columns: 2

Enter first matrix:1

2

3

4

Enter second matrix:

5

6

7

8

Sum of matrices:

6 8

10 12

2.REVERSAL OF AN ARRAY :

```
#include <stdio.h>

int main()
{
   int n, i;

   printf("Enter the size of array: ");
   scanf("%d", &n);

   int a[n];

   printf("Enter the elements:\n");
   for(i = 0; i < n; i++)
   {
      scanf("%d", &a[i]);
   }

   printf("Reversed array:\n");
```

```
    for(i = n - 1; i >= 0; i--)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT :

Enter the size of array: 7

Enter the elements:

25

45

87

56

3

89

65

Reversed array:

65 89 3 56 87 45 25

3.TO FIND DUPLICATES IN AN ARRAY

```
#include <stdio.h>

int main()
{
    int n, i, j, count = 0;

    printf("Enter the size of array: ");
    scanf("%d", &n);
```

```
    int a[n];

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[i] == a[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total number of duplicate elements = %d", count);

    return 0;
}
```

OUTPUT :

Enter the size of array: 5

Enter the elements:

10 20 20 30 40

Total number of duplicate elements = 1

4.UNIQUE ELEMENTS IN AN ARRAY

```c
#include <stdio.h>

int main()
{
    int n, i, j, count;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    int a[n];

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Unique elements are:\n");

    for(i = 0; i < n; i++)
    {
        count = 0;

        for(j = 0; j < n; j++)
        {
            if(a[i] == a[j])
            {
                count++;
            }
        }

        if(count == 1)
```

```
        {
            printf("%d ", a[i]);
        }
    }

    return 0;
}
```

OUTPUT :

Enter the size of array: 5

Enter the elements:

10 20 20 30 30

Unique elements are:

10

5.INSERTING AN ELEMENT IN SORTED ARRAY

```
#include <stdio.h>

int main()
{
    int n, i, element, pos;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    int a[n + 1];

    printf("Enter the sorted array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
```

```
    }

    printf("Enter the element to insert: ");
    scanf("%d", &element);

    pos = n;

    for(i = 0; i < n; i++)
    {
        if(element < a[i])
        {
            pos = i;
            break;
        }
    }

    for(i = n; i > pos; i--)
    {
        a[i] = a[i - 1];
    }

    a[pos] = element;

    printf("Array after insertion:\n");

    for(i = 0; i <= n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT :

Enter the size of array: 5

Enter the sorted array elements:

25 65 89 98 99

Enter the element to insert: 87

Array after insertion:

25 65 87 89 98 99

6.INSERTING AN ELEMENT IN UNSORTED ARRAY

```
#include <stdio.h>

int main()
{
    int n, i, element, pos;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    int a[n + 1];

    printf("Enter the array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter the element to insert: ");
    scanf("%d", &element);

    printf("Enter the position: ");
    scanf("%d", &pos);
```

```
    for(i = n; i >= pos; i--)
    {
        a[i] = a[i - 1];
    }

    a[pos - 1] = element;

    printf("Array after insertion:\n");

    for(i = 0; i <= n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT :

Enter the size of array: 5

Enter the array elements:

89 56 21 45 32

Enter the element to insert: 57

Enter the position: 3

Array after insertion:

89 56 57 21 45 32

7.MATRIX MULTIPLICATION

```
#include <stdio.h>

int main()
{
```

```
int r1, c1, r2, c2, i, j, k;

printf("Enter rows and columns of first matrix: ");
scanf("%d %d", &r1, &c1);

printf("Enter rows and columns of second matrix: ");
scanf("%d %d", &r2, &c2);

if(c1 != r2)
{
    printf("Matrix multiplication is not possible.");
    return 0;
}

int a[r1][c1];
int b[r2][c2];
int result[r1][c2];

printf("Enter first matrix:\n");
for(i = 0; i < r1; i++)
{
    for(j = 0; j < c1; j++)
    {
        scanf("%d", &a[i][j]);
    }
}

printf("Enter second matrix:\n");
for(i = 0; i < r2; i++)
{
    for(j = 0; j < c2; j++)
    {
```

```
            scanf("%d", &b[i][j]);
        }
    }

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            result[i][j] = 0;

            for(k = 0; k < c1; k++)
            {
                result[i][j] = result[i][j] + a[i][k] * b[k][j];
            }
        }
    }

    printf("Result of matrix multiplication:\n");

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            printf("%d ", result[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT :

Enter rows and columns of first matrix: 2 3

Enter rows and columns of second matrix: 3 2

Enter first matrix:

23 45 56 1 2 3

Enter second matrix:

1 2 3 56 45 23

Result of matrix multiplication:

2678 3854

142 183

8.MATRIX TRANSPOSE

```
#include <stdio.h>

int main()
{
    int rows, cols, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &rows);

    printf("Enter number of columns: ");
    scanf("%d", &cols);

    int a[rows][cols];
    int transpose[cols][rows];

    printf("Enter the matrix elements:\n");

    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
```

```
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            transpose[j][i] = a[i][j];
        }
    }

    printf("Transpose of the matrix:\n");

    for(i = 0; i < cols; i++)
    {
        for(j = 0; j < rows; j++)
        {
            printf("%d ", transpose[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT :

```
Enter number of rows: 2
Enter number of columns: 3
Enter the matrix elements:
1 2 3 4 5 6
Transpose of the matrix:
```

1 4

2 5

3 6

9.RIGHT DIAGONAL SUM

```
#include <stdio.h>

int main()
{
    int n, i, j, sum = 0;

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    int a[n][n];

    printf("Enter the matrix elements:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        sum = sum + a[i][n - 1 - i];
    }
```

```
    printf("Sum of right diagonal = %d", sum);

    return 0;
}
```

OUTPUT :

Enter the size of matrix: 2

Enter the matrix elements:

2 5 6 8

Sum of right diagonal = 11

10.CHECK IDENTITY MATRIX

```
#include <stdio.h>

int main()
{
    int n, i, j, flag = 1;

    printf("Enter the size of matrix: ");
    scanf("%d", &n);

    int a[n][n];

    printf("Enter the matrix elements:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
```

```
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j)
            {
                if(a[i][j] != 1)
                {
                    flag = 0;
                }
            }
            else
            {
                if(a[i][j] != 0)
                {
                    flag = 0;
                }
            }
        }
    }

    if(flag == 1)
        printf("The matrix is an Identity Matrix.");
    else
        printf("The matrix is not an Identity Matrix.");

    return 0;
}
```

OUTPUT :

CASE 1

Enter the size of matrix: 3

Enter the matrix elements:

1 0 0 0 1 0 0 0 1

The matrix is an Identity Matrix**.**

CASE 2

Enter the size of matrix: 3

Enter the matrix elements:

1 2 3 4 5 6 7 8 9

The matrix is not an Identity Matrix**.**

11.COUNT VOWELS, CONSONANTS, DIGITS, WHITE SPACES

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;
    int vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'a' && str[i] <= 'z') ||
           (str[i] >= 'A' && str[i] <= 'Z'))
        {
            if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
               str[i] == 'o' || str[i] == 'u' ||
               str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
               str[i] == 'O' || str[i] == 'U')
            {
```

```
            vowels++;
          }
          else
          {
            consonants++;
          }
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
          digits++;
        }
        else if(str[i] == ' ')
        {
          spaces++;
        }
      }

      printf("Vowels = %d\n", vowels);
      printf("Consonants = %d\n", consonants);
      printf("Digits = %d\n", digits);
      printf("White spaces = %d\n", spaces);

      return 0;
    }
```

OUTPUT :

Enter a string: harish 2008

Vowels = 2

Consonants = 4

Digits = 4

White spaces = 1

12.FREQUENCY OF A CHARACTER IN A STRING

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, j, count;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ' ' || str[i] == '\n')
            continue;

        count = 1;

        for(j = i + 1; str[j] != '\0'; j++)
        {
            if(str[i] == str[j])
            {
                count++;
            }
        }

        // Check whether character was already counted
        for(j = 0; j < i; j++)
        {
            if(str[i] == str[j])
            {
                count = 0;
                break;
```

```
        }
      }

      if(count != 0)
      {
        printf("%c = %d\n", str[i], count);
      }
  }

  return 0;
}
```

OUTPUT :

Enter a string: HARISH

H = 2

A = 1

R = 1

I = 1

S = 1

13.COUNT WORDS IN A STRING

```
#include <stdio.h>

int main()
{
  char str[100];
  int i, words = 0;

  printf("Enter a string: ");
  fgets(str, sizeof(str), stdin);

  for(i = 0; str[i] != '\0'; i++)
```

```
    {
        if(str[i] != ' ' && (str[i + 1] == ' ' || str[i + 1] == '\n'))
        {
            words++;
        }
    }

    printf("Number of words = %d", words);

    return 0;
}
```

OUTPUT :

Enter a string: MY NAME IS HARISH

Number of words = 4

14.SORT STRING ARRAY

```
#include <stdio.h>
#include <string.h>

int main()
{
    int n, i, j;
    char temp[100];

    printf("Enter number of strings: ");
    scanf("%d", &n);

    char str[n][100];

    printf("Enter the strings:\n");
```

```
    for(i = 0; i < n; i++)
    {
        scanf("%s", str[i]);
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Strings in alphabetical order:\n");

    for(i = 0; i < n; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

OUTPUT :

Enter number of strings: 3

Enter the strings:

AJAY SIVA GANESH

Strings in alphabetical order:

AJAY

GANESH

SIVA

15.TOGGLE CASE OF A SENTENCE

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        else if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
    }

    printf("Toggled sentence: %s", str);

    return 0;
}
```

OUTPUT :

Enter a sentence: Darshath is a good BOY

Toggled sentence: dARSHATH IS A GOOD boy

****